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RESEARCH ARTICLE

NATYA-AI: A Cultural AI Framework for Multimodal Interpretation of Bharatanatyam

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ABSTRACT Bharatanatyam is a classical Indian dance form that conveys structured narratives through a combination of symbolic hand gestures (mudras), facial expressions (bhavas), and lyrical content. In this research, we present **NATYA-AI**, an engineering pipeline for multimodal semantic interpretation of Bharatanatyam performances from video data. The system integrates computer vision models, including YOLO and MediaPipe, for hand gesture detection, and convolutional neural networks for facial emotion classification. It employs Whisper ASR for transcription of multilingual devotional lyrics. These visual and auditory components are processed and semantically fused using the Gemini large language model to generate context-aware narrative summaries. The output includes both textual and audio representations of the interpreted performance, enhancing usability for educational, archival, and accessibility-focused applications. NATYA-AI is implemented as a standalone Python tool, with open-source code and example data publicly available. Experimental evaluations conducted under varying lighting and camera conditions confirm the system's ability to reliably recognize gestures, emotional expressions, and lyrical content. The results demonstrate the feasibility of applying multimodal AI to the semantic interpretation of classical Indian dance, underscoring the potential of **NATYA-AI** not only for post-performance interpretation but also as a foundation for future real-time analysis and interactive cultural learning. Unlike prior vision-only or audio-only systems, NATYA-AI performs cross-modal fusion of symbolic gestures, emotional expressions, and lyrical meaning to generate culturally coherent narratives.

INDEX TERMS Large language models, Navarasa, whisper, Gemini, mudra recognition, convolutional neural network.

I. INTRODUCTION

NATYA-AI is a multimodal modular AI pipeline that was built to semantically understand Bharatanatyam performances through the analysis of gestures, facial expressions, and lyrical meaning. It integrates vision-based emotion and mudra classification to the analysis, automatic speech recognition, and large language model reasoning to produce narrative explanations for classical Indian dance. The system is tailored for educational assistance, preservation of culture, and audience accessibility.

Bharatanatyam, an ancient and sacred Indian classical dance, is an emotionally charged narrative medium. Based on the treatise *Natya Shastra*, it weaves subtle hand gestures (mudras), facial expressions (bhavas), rhythmic foot movements, and elegant body positions to depict mythological stories, spiritual motifs, and emotional states. This largely symbolic and visually dense language forms the core vocabulary of Bharatanatyam, often synchronized with classical compositions in Tamil, Sanskrit, Telugu, or Kannada that lend lyrical and rhythmic structure to performances.

Despite its artistic depth and cultural sophistication, Bharatanatyam remains largely inaccessible to untrained viewers. Its symbolic gestures and lyrical metaphors require years of guided learning to interpret. The scarcity of annotated datasets, standard learning pathways, and language barriers further impede broader engagement. These challenges affect learners, researchers, and international audiences alike, limiting interpretive access to this expressive art form.

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A. RESEARCH MOTIVATION AND QUESTION

Recent advancements in artificial intelligence (AI), particularly in multimodal learning involving vision, speech, and language, have shown promising capabilities in interpreting complex human communication. However, their application to traditional performance arts like Bharatanatyam is underexplored. Bharatanatyam's unique fusion of visual, linguistic, and emotional elements presents a compelling use case for AI-based interpretation.

This paper addresses the following research question:

> ***How can multimodal AI methods such as computer vision, speech recognition, and large language models (LLMs) be combined to automatically recognize and describe Bharatanatyam performances from video?***

To explore this question, we present **NATYA-AI** - a unified multimodal system for semantic interpretation of Bharatanatyam. NATYA-AI integrates computer vision for detecting gestures and expressions, Whisper ASR for multilingual lyric transcription, and LLMs for transliteration and semantic analysis. These components are contextually fused to generate comprehensive narrative storylines in both textual and audio formats. The system is designed not only for performance analysis, but also for educational support, cultural preservation, and audience accessibility.

B. MAJOR CONTRIBUTIONS

The novelty of NATYA-AI is not applying existing AI models in isolation, but integrating them, appending proper context, and in culturally related and appropriate contexts within Bharatanatyam. The most important contributions of this work are below:

- Multimodal fusion pipeline: We have put forth a cohesive architecture that moves beyond traditional unimodal classification tasks by unifying gesture, facial expressions, and lyric analysis to create coherent, contextually relevant narratives.
- Cultural-semantic grounding: Recognition outputs are linked to canonical Bharatanatyam concepts, such as the Navarasas and the mudra lexicon, which ensures that performers maintain validity of interpretations and are grounded in the principles and philosophies handed down as pedagogical narratives.
- LLM-based narrative generation: we used a large language model (Gemini) applying suitable prompting techniques to help us construct narratives from synthesizing multimodal signals in an effort to produce strong cultural context narratives as performance interpretations.
- Outputs based on accessibility: NATYA-AI provides the potential for text-based and audio-based dual formats for interpretive outputs that maximize accessibility for learners, teachers, and blind and visually-impaired audiences alike.
- Cultural AI contribution: NATYA-AI represents one of the first examples of Cultural AI applied to Indian classical dance, and we offer a framework for further development and research in a way that helps preserve digital heritage, and advance research in the computational arts.

While the individual models used in NATYA-AI (YOLO, MediaPipe, CNNs, Whisper ASR, and Gemini LLM) are established, their integration within a culturally grounded semantic framework defines the novelty of this work. NATYA-AI is the first system to achieve end-to-end fusion of gestures, facial emotions, and lyrical meaning for Bharatanatyam, generating coherent narrative interpretations rather than feature-level alignment. By embedding domain knowledge through curated mappings of mudras and Navarasa, and producing dual-format outputs (text and audio) for accessibility, the system demonstrates a unique application of multimodal AI in Cultural Informatics, advancing the interpretive understanding and preservation of intangible heritage.

C. IMPORTANCE OF CULTURAL AI

This research contributes to the emerging domain of *Cultural AI*, where AI is applied to safeguard, interpret, and promote traditional knowledge systems. Bharatanatyam, with its symbolic and rule-based performance language, lends itself well to computational modeling. By translating gestures, expressions, and lyrics into semantically meaningful narratives, NATYA-AI opens new pathways for engaging with cultural heritage through digital means.

The proposed system provides utility across domains — enabling students **to learn through feedback**, supporting scholars in semantic interpretation, and helping diverse audiences understand the emotional and spiritual dimensions of Bharatanatyam. It aligns with broader efforts to democratize cultural access using intelligent systems.

D. ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs)

The work aligns with several United Nations Sustainable Development Goals:

- **SDG 4: Quality Education** — Providing inclusive and accessible pathways to understand Bharatanatyam, independent of linguistic or cultural background.
- **SDG 11: Sustainable Cities and Communities** — Contributing to the preservation and continuity of intangible cultural heritage.
- **SDG 9: Industry, Innovation, and Infrastructure** — Developing AI infrastructure for digital humanities and creative cultural applications.
- **SDG 16: Peace, Justice, and Strong Institutions** — Fostering intercultural understanding through access to cultural narratives.

E. TECHNICAL CONTEXT

Recent advances in deep learning have driven significant improvements in image recognition, speech transcription, and narrative generation. Techniques such as keyframe extraction using clustering and genetic algorithms [1], [2], speech transcription via Whisper [3], [4], and facial expression classification using CNNs [5], [6] provide the foundational capabilities required for this work. Prior research on LLM-based storytelling [7], [8] has shown the promise of generative models for synthesizing narrative from multimodal data.

Building on this foundation, NATYA-AI unifies gesture classification, emotion recognition, and lyric interpretation within a single semantic pipeline. To our knowledge, it is the first system to generate synchronized, context-aware narratives for Bharatanatyam by aligning visual, auditory, and symbolic elements. In doing so, it contributes both a technical advancement in multimodal AI and a cultural innovation in digital heritage interpretation.

II. RELATED WORKS

Recent advances in AI and machine learning (ML) have enabled significant progress in multimedia understanding, particularly in video analysis, speech recognition, facial expression classification, and natural language processing. These capabilities form the foundation for developing intelligent systems that can interpret multimodal performances such as Bharatanatyam.

This section reviews existing literature relevant to the proposed system, focusing on four key components: (i) video frame extraction and gesture recognition, (ii) Lyrics extraction, (iii) lyrics description, (iv) Facial Expression Classification, and (iv) narrative generation using LLMs. Each of these areas contributes to the broader goal of automatic interpretation of expressive human activities.

In particular, prior work has explored various frame extraction techniques using clustering and deep learning for efficient video summarization. Similarly, speech transcription models such as Whisper have advanced multilingual ASR, making classical song lyrics more accessible. Deep learning methods, especially CNNs, have demonstrated strong performance in facial expression recognition, including applications to traditional Indian dances. Finally, the use of LLMs such as GPT and Gemini has shown promise in generating context-aware narratives across domains ranging from education to cultural storytelling.

The following subsections elaborate on relevant contributions in each of these domains and position our approach in the context of this growing body of work.

A. VIDEO FRAME EXTRACTION

Efficient extraction of key frames from video is essential for summarization, indexing, and downstream multimedia analysis tasks. Several techniques have been proposed to segment videos and identify representative frames that preserve contextual and semantic information.

Kumar et al. [9] proposed a method for converting video to frames using MATLAB, particularly focusing on TV news content. Their approach employed shot detection techniques such as histogram comparison, statistical difference, and motion vector analysis to identify significant transitions. They demonstrated that their method could improve data processing and storage efficiency by accurately segmenting the video into key segments.

Mithlesh et al. [10] also explored video frame extraction using MATLAB to support indexing and summarization. Their approach used algorithmic criteria including frame rate analysis and aspect ratio calculations to extract key frames from online video databases. The study further compared different video formats and frame properties to assess extraction efficiency, confirming its usefulness in applications like video retrieval and surveillance.

Zhang et al. [1] developed a clustering-based method for key-frame extraction from motion capture data. By clustering adjacent frames using similarity distance and applying the ISODATA algorithm, the method adaptively selected representative frames without requiring user-defined thresholds. Their evaluation, based on motion reconstruction and error minimization, showed improved reliability in summarizing motion data.

Rajan et al. [2] introduced a genetic algorithm-based approach for key-frame extraction in surveillance video. Their method optimized frame selection using a fitness function and elitist survivor selection, and was tested on datasets such as VSUMM, SumMe, and real-world surveillance videos. Compared to differential evolution and deep learning approaches, the genetic algorithm demonstrated improved performance by reducing redundancy while preserving critical video content.

These works collectively inform the visual processing pipeline used in our system, where representative frames are selected for gesture and expression recognition in Bharatanatyam performance videos.

B. LYRICS EXTRACTION

Automatic speech recognition (ASR) plays a critical role in processing audio components of multimodal content. Whisper, developed by OpenAI, has emerged as a prominent model in multilingual transcription due to its generalizability and robustness.

Khairani et al. [4] incorporated Whisper and the Google Translate API into an Arabic learning platform, enabling real-time speech-to-text transcription and translation. Their system also employed CNN-based pronunciation training, achieving an accuracy of 95.52%. Machine learning techniques were used to optimize model parameters and enhance both transcription performance and user engagement, demonstrating the potential of ASR in language learning applications.

Radford et al. [3] introduced Whisper as a large-scale, weakly supervised ASR model trained on 680,000 hours of multilingual data. The system utilizes a transformer-based encoder-decoder architecture and achieves zero-shot performance across various languages and tasks. Their work illustrates that pretraining on diverse datasets can significantly enhance transcription quality while reducing the need for task-specific fine-tuning. Notably, Whisper demonstrated performance approaching human-level accuracy across multiple speech benchmarks.

Surabhi et al. [11] performed a systematic overview of diverse TTS and STT models, from the old rule-based systems to the latest state-of-the-art deep learning models. Their research classifies models as per training methods, architectural styles, and application fields and throws light on how these technologies aid virtual assistants, assistive devices, and multilingual communication. This thorough examination brings out the advances and ongoing challenges in attaining natural, precise, and context-sensitive speech-language interaction, providing a rich resource for future research and development on voice-based systems.

Dhankar et al. [12] compared traditional ASR techniques with deep learning approaches. They evaluated CNN-based models against the CMU Sphinx system, which is based on Gaussian Mixture Model–Hidden Markov Model (GMM-HMM) architecture. Their findings showed that deep learning models outperformed traditional ASR systems in terms of accuracy, especially when processing large datasets. This highlights the scalability and improved generalization of modern ASR techniques in speech-related tasks.

These studies support the adoption of Whisper in our system, where accurate transcription of classical Bharatanatyam song lyrics—often in Tamil or Sanskrit—is essential for subsequent semantic analysis.

C. LYRICS DESCRIPTION

The semantic interpretation of lyrics is crucial for understanding cultural and emotional contexts in performance arts. Recent research has explored the role of LLMs in generating narrative and thematic content from textual inputs.

Johnson-Bey et al. [7] investigated the application of ChatGPT for generating coherent narratives within simulated environments. Their system utilized natural language prompts to generate structured data, including businesses, occupations, and character attributes, which were then integrated into the Neighborly simulation platform. Although the generated content was useful, the study also addressed challenges related to narrative consistency and content management. Their findings underscore the capacity of LLMs to automate storytelling and content generation, particularly in interactive and culturally rich environments.

Sentiment analysis tools have also been extensively researched in multimedia feedback settings. Tetteh et al. [13] gave a rich overview of the sentiment analysis tools applied in analyzing movie reviews. They included known models like TextBlob and VADER, enumerating their efficiency in measuring public opinion from text sources. While focused on movie content, their results are helpful for sentiment-conscious textual analysis in cultural fields like Bharatanatyam, particularly when interpreting lyrical or narrative material. Veena et al. [14] introduced a more advanced sentiment analysis method that combines VADER and dependency parsing in order to catch sentence structure and context more effectively. Their model also considers using BERT for more fine-grained sentiment classification. Although the research is conducted for general text data, the methods are easily usable in semantic analysis of traditional lyrics in which syntactic dependency and emotional tone are usually embedded in poems. This blended approach aids enhanced polarity detection in our system's lyrical analysis pipeline.

These insights support the use of LLMs in our project, where the goal is to interpret Bharatanatyam lyrics and generate meaningful storylines that reflect both emotional tone and cultural symbolism.

D. MUDRA IDENTIFICATION

Some researchers have searched to apply machine learning and deep learning algorithms for the detection and classification of Bharatanatyam mudras.

Haridas et al. [15] suggested a revised deep learning model to detect and classify Bharatanatyam mudras, showing prospective accuracy by utilizing domain-specific data sets. Santhosh et al. [16] examined the classification of Asamyukta mudras via handcrafted features as well as pre-trained deep models. Their hybrid method emphasized the power of mashing up traditional and contemporary methods for better performance. Sadhana et al. [17] used meta-learning methods in addition to deep learning for overcoming the problem of having scarce labeled data so that efficient mudra recognition is possible even with limited training examples.

Classic machine learning techniques have also been utilized; Varsha et al. [18]employed Histogram of Oriented Gradients (HOG) descriptors along with a Support Vector Machine (SVM) classifier to detect hand gestures, providing a low-weight yet efficient solution. Naaz et al. [19] proposed an end-to-end deep learning model that is proficient in classifying different Bharatanatyam mudras, particularly in situations with intricate hand overlaps. Kumar et al. [20] proposed a progressive attention-based model with wavelet channels for including fine-grained pose information, enhancing recognition between two adjacent dance frames.

Malavath et al. [21] combined ancient dance knowledge and AI through the application of deep learning networks in hand mudra recognition as per the Natya Shastra. Their system reported good performance in video-based gesture classification. In the above context, Jisha et al. [22] also emphasized systematic tuning of CNNs for mudra classification, demonstrating that ideal model calibration greatly enhances accuracy. Finally, real-time hand tracking systems such as MediaPipe Hands [23] have been useful in gesture preprocessing, while systems such as Yaseen et al. [24] coupled with MediaPipe, Inception-V3, and LSTM have set the stage for dynamic gesture recognition suitable for classical dance applications.

Aloysius et al. [25] investigated Transformer-based models for sign language translation and recognition. Their research emphasized the significance of spatial human joint relations in understanding gestures. By integrating relative positional encoding into self-attention within Transformers, the model succeeded in learning fine-grained motions important for complex sign recognition. The method attained better performance on benchmark sign language datasets and provides significant contributions towards recognition tasks with structured body movement such as Bharatanatyam dance gestures.

E. CLASSIFICATION OF FACIAL EXPRESSION

Facial expressions play a critical role in Bharatanatyam, particularly in conveying emotional states through the Navarasa framework. Recent research has demonstrated the effectiveness of deep learning techniques, particularly CNNs, for facial expression classification in traditional Indian dance forms.

Selvi et al. [5] developed a deep learning-based system for classifying Navarasa expressions in Kathakali using image processing methods. They constructed a dataset comprising 650 images representing nine distinct emotional expressions, curated with assistance from expert Kathakali artists. A CNN-based classifier achieved an initial accuracy of 90%, which improved to 93% with the integration of a fuzzy preprocessing algorithm. This study demonstrated the feasibility of applying AI techniques for accurate emotion recognition in classical Indian dance, contributing to both cultural preservation and accessibility.

Reshma et al. [6] proposed a real-time annotation system for Navarasa expressions in video using CNNs. The model was trained on a dataset of labeled facial expressions and showed high recognition accuracy in live video streams. Their results indicate that deep learning models can successfully interpret facial cues in dynamic performance settings, offering potential applications in both archival and educational contexts.

Baggam et al. [26] conducted a comparative study on facial recognition methods using the ORL face dataset, evaluating machine learning (ML), deep learning (DL), and Bayesian approaches. Their findings confirmed that deep learning methods consistently outperformed traditional techniques in modeling complex facial features, making them well-suited for high-precision tasks such as emotion classification in classical dance.

These studies collectively validate the adoption of CNN-based architectures for facial expression recognition in our system, enabling accurate identification of emotional cues essential to understanding the narrative depth of Bharatanatyam performances.

F. LARGE LANGUAGE MODELS (LLMs)

LLMs have demonstrated substantial capabilities in generating coherent, context-aware text across a range of domains, including education, storytelling, and intelligent system design.

Chen et al. [8] investigated the integration of ChatGPT with mind mapping techniques to support children's storytelling and comprehension. In their study, ChatGPT was used to generate age-appropriate stories, construct structured mind maps, and design test questions to assess learners' understanding. A quasi-experimental design evaluated the effectiveness of AI-assisted storytelling tools, and results indicated improvements in narrative coherence and conceptual clarity. This research highlights the educational potential of LLMs for adaptive and creative learning environments.

Sriram et al. [27] proposed a framework that employs locally hosted LLMs for secure, on-device task automation. Their system utilizes prompt-based agent classification to enable decision-making and natural language understanding without depending on external cloud services. This approach ensures data privacy while maintaining functional autonomy, making it well-suited for sensitive or culturally specific applications.

In wider image classification tasks as well, ensemble-based deep CNN methods have surpassed traditional techniques [28], further demonstrating the robustness and adaptability of CNNs. Building on these developments, our system uses CNNs for face recognition and then combines them with large language models to meaningfully understand and narrate Bharatanatyam performances.

These studies support the incorporation of LLMs in our system for semantic interpretation of Bharatanatyam lyrics and narrative generation. By integrating models such as Gemini, we aim to produce contextually accurate storylines while preserving user privacy and ensuring efficient on-device performance.

G. INDIC LANGUAGE TRANSLATION AND PREPROCESSING

Effective translation and preprocessing of Indic languages, particularly Tamil, are essential for semantically accurate interpretation of Bharatanatyam lyrics. Recent advancements in multilingual language models and preprocessing toolkits have significantly improved performance on low-resource and morphologically complex languages.

Conneau et al. [29] introduced XLM-R, a self-supervised cross-lingual model trained on large-scale multilingual corpora covering over 100 languages, including Tamil. The model demonstrated strong zero-shot translation capabilities without the need for language-specific fine-tuning. Its ability to retain syntactic and semantic features across languages makes it well-suited for interpreting Bharatanatyam lyrics, where cultural nuance and grammatical complexity are key.

Liang et al. [30] presented LLaMA 2, a family of open-source foundation models optimized for multilingual understanding and generation tasks. Designed to perform well on languages with limited training data, LLaMA 2 models showed improved contextual comprehension and semantic preservation. These characteristics are particularly beneficial for translating Indian classical song lyrics that often involve poetic structures and metaphorical language.

Adelani et al. [31] developed MasakhaNER, a multilingual named entity recognition dataset and model for low-resource languages, including Tamil. Their work emphasized the importance of preserving linguistic and cultural context in tasks such as translation and entity recognition. This is directly relevant to our system, which must interpret Bharatanatyam lyrics containing references to mythological figures and culturally specific terminology.

Kakwani et al. [32] introduced the IndicNLP Suite, a collection of preprocessing tools tailored for Indian languages. These tools provide functionalities such as tokenization, script normalization, and sentence segmentation. Their toolkit addresses challenges inherent in handling Indic scripts and is highly applicable in preparing Tamil lyrics for downstream processing by LLMs.

Lewis et al. [33] proposed BART, a sequence-to-sequence model based on a denoising autoencoder architecture that integrates both bidirectional and autoregressive transformers. BART has shown strong performance in translation and summarization tasks, serving as the foundation for many contemporary LLM-based generation models. Its capabilities in preserving linguistic structure and generating coherent output make it a relevant model for translating and interpreting classical lyrical compositions.

Collectively, these studies support the use of multilingual models and preprocessing frameworks in our pipeline to ensure accurate, context-sensitive translation and semantic analysis of Bharatanatyam lyrics.

H. SUMMARY AND RESEARCH GAPS

The reviewed literature highlights significant progress in key areas relevant to this study, including video frame extraction, facial expression classification, speech recognition, LLMs, and Indic language processing. Frame extraction techniques have been effectively used for tasks such as surveillance and video summarization, while deep learning models, particularly CNNs, have proven successful in recognizing facial expressions within classical Indian dance forms like Kathakali. Similarly, speech-to-text models such as Whisper have demonstrated strong performance in multilingual transcription, and LLMs like ChatGPT and LLaMA 2 have shown potential in generating coherent narratives from textual input. Additionally, preprocessing tools like IndicNLP Suite and multilingual translation models including XLM-R and BART have improved the handling of Tamil and other low-resource languages.

Despite these advancements, certain research gaps remain. Most existing systems focus on individual modalities and do not offer an integrated approach that combines gestures, facial expressions, and lyrics for holistic interpretation. There is also limited research on applying LLMs to culturally specific domains such as Bharatanatyam, where the content is often symbolic and rooted in mythology. Moreover, while tools exist for speech transcription and translation, their application to classical Tamil lyrics—characterized by poetic and metaphorical language—has not been widely studied. Another gap is the lack of systems that convert these multimodal inputs into coherent storylines suitable for learners and cultural scholars. Furthermore, few studies have presented end-to-end systems with real-world applications aimed at education, accessibility, or cultural preservation.

This study addresses these gaps by proposing a unified system that combines visual and linguistic cues to automatically interpret Bharatanatyam performances and generate narrative outputs in both text and audio formats.

III. METHODOLOGY

The proposed system, titled **NATYA-AI**, is designed to enable structured interpretation of Bharatanatyam—a highly expressive Indian classical dance form—through a fusion of contemporary AI technologies. Bharatanatyam performances are composed of intricate hand gestures (mudras), facial expressions (bhavas), and poetic lyrics, all of which work in concert to convey nuanced emotional and narrative content. However, for learners, hobbyists, or audiences unfamiliar with its symbolic vocabulary, decoding the performance's meaning remains a substantial challenge.

NATYA-AI addresses this gap by integrating computer vision, natural language processing (NLP), and LLMs to analyze multimodal data extracted from dance videos. The system processes visual elements to detect and classify mudras and facial expressions using YOLO, MediaPipe, and CNNs. Simultaneously, it transcribes and semantically interprets the accompanying lyrics—often in Tamil or other classical languages—using Whisper ASR and the Gemini LLM. These independent streams of information are then contextually fused to generate a coherent narrative that reflects the thematic and emotional arc of the performance.

The output of NATYA-AI is not limited to textual interpretation; it also supports audio narration, enabling greater accessibility and engagement for diverse users, including those who benefit from auditory reinforcement or require assistive technologies. The system's layered design also allows for integration with sign language interpretation tools, further broadening its accessibility across learning and cultural contexts.

To validate the effectiveness of the proposed framework, a comparative system-level evaluation is conducted against existing approaches. Table 1 presents a comparative analysis, highlighting the unique contributions and multimodal depth offered by NATYA-AI. In parallel, Figure 1 illustrates the architectural workflow, beginning from input

TABLE 1. System-level comparison of proposed framework with prior research.

Component	Existing Works	Proposed System
Mudra Recognition	CNNs on static images; limited to single-hand gestures	YOLO + MediaPipe + MobileNetV2 for detecting both single and double-hand mudras
Facial Expression Analysis	Navarasa detection via basic CNNs in Bharatanatyam/Kathakali	Custom CNN trained on 9 expressions with semantic and temporal mapping
Lyrics Transcription	Whisper/Google ASR used in isolation; limited in handling poetic Tamil lyrics	Whisper Large v2 with audio filtering and domain-specific preprocessing
Lyrics Interpretation	Basic translation using generic NLP tools	Gemini LLM for semantic interpretation and Bhakti-based contextual meaning
Storyline Generation	Rare or non-existent in classical dance domains	LLM-driven narrative fusion of gestures, expressions, and lyrics
Multimodal Fusion	Mostly focused on one or two modalities	Unified architecture combining vision, audio, and text inputs with temporal alignment
Accessibility Output	Classification results only; not user-oriented	PDF + Audio output generation for inclusive educational and research use
Cultural AI Positioning	Implicit or not discussed in technical papers	Explicitly aligned with SDGs and digital heritage preservation goals

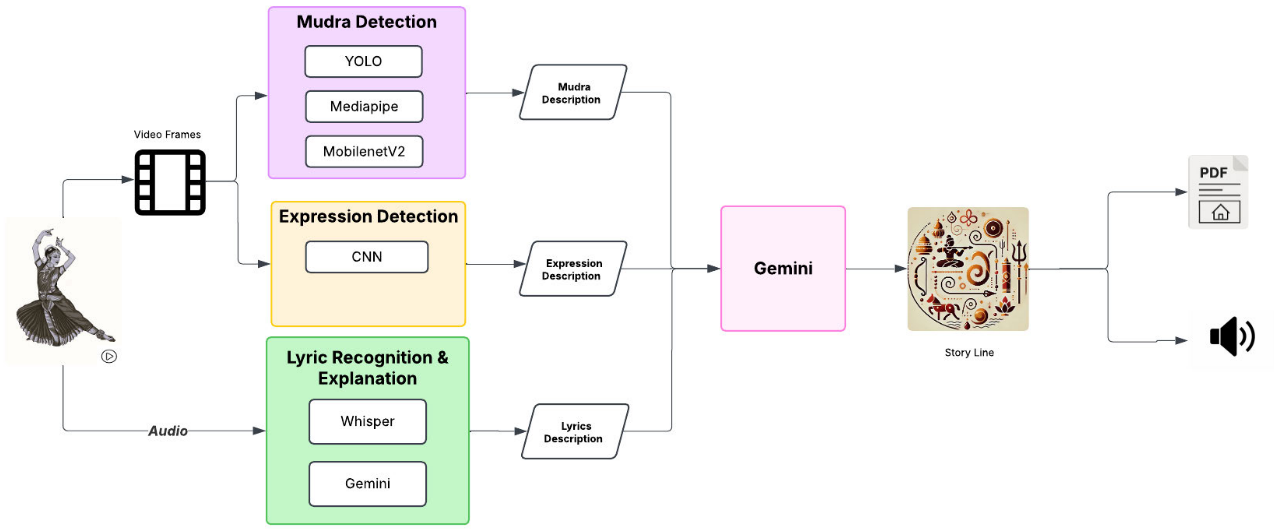

FIGURE 1. System architecture of NATYA-AI, illustrating the multimodal pipeline: gesture and facial expression detection, lyrics transcription (Whisper ASR), and narrative generation (Gemini LLM).

acquisition (video, audio) and extending through sequential processing modules—gesture and emotion classification, lyric transcription, semantic reasoning, and narrative generation.

Together, the system diagram and comparative analysis illustrate the performance gains of NATYA-AI over prior static or unimodal interpretive systems. These advantages are particularly evident in terms of contextual fidelity, narrative coherence, and cultural relevance, which are critical when interpreting art forms that rely heavily on symbolic abstraction.

By enabling interpretable and context-aware storytelling, NATYA-AI serves as a valuable tool for learners, educators, and researchers alike. It facilitates a more inclusive engagement with Bharatanatyam by revealing the intertwined meanings of movement, expression, and music in a comprehensible and structured form. The further subsections details each module of the architecture.

A. VIDEO ACQUISITION AND PREPROCESSING

The input of the proposed system is comprised of classical Bharatanatyam dance video performances available on open video sharing websites called YouTube. The yt-dlp utility was used to download the videos, and it is a reliable utility for downloading. The short video that was downloaded is from https://www.youtube.com/shorts/uexw920VSdA3, and resulted in a.webm file. After downloading the YouTube videos, it was loaded into a Python environment using class *moviepy.editor.VideoFileClip*, which gave access to audio and video data of the short Bharatanatyam video performance. From this stage of the project, downstream processing will consist of: extracting each frame of the video for gesture recognition, and separating audio from video for lyrics transcription.

B. VIDEO FRAME EXTRACTION FOR GESTURE AND EXPRESSION ANALYSIS

In the proposed system, video input is examined, which enables the system to support facial expression and hand gesture (mudra) recognition tasks. Once a video is loaded using *moviepy*, the video is split into frames at fixed intervals and viewed as a sequence of frames that can be examined one by one. The individual frames can then be used as input to two classification modules, one for classifying the Bharatanatyam mudras and another for classifying the facial expressions associated with the Navarasas. Using the video frames for both classification modules allows the system to take advantage of both the physical gesture of the hands (mudras) and the emotional expression shown on the face–together providing a more complete and culturally rich understanding of a Bharatanatyam performance.

C. AUDIO EXTRACTION AND ENHANCEMENT

The audio was extracted from the video file using the *moviepy* library and saved in.mp3 format. The underlying audio file format a 16 kHz mono-channel.wav file was used to import into the transcription model that because the model does not accept.mp3 files. A high-pass and low-pass filter was used to enhance the audio and reduce some background noise during the last process. Once the audio was enhanced, the transcription of the audio was completed using the Whisper large-v2 model created by OpenAI. The audio transcription process was important to help understand the lyrical contextualization of the Bharatanatyam performance and can be matched with recognized gestures and expressions to create an interpretable story.

D. MUDRA DETECTION

In our earlier work [34] we established a system to recognize Bharatanatyam mudras through the use of YOLO for detecting hands, MediaPipe Hand Landmark detection to obtain skeletal keypoints, and a MobileNetV2-based classifier for classifying mudras. Images or frames from videos, sourced from our publicly released Mudra Dataset [35], were analyzed to detect single-hand and double-hand mudras. Every classified mudra was then labeled with its corresponding cultural description from a curated corpus, facilitating the digital preservation and educational sharing of Bharatanatyam hand gestures. The architecture of the Mudra detection framework is shown in Figure 2. An Excel sheet has been prepared, as shown in Table 2, that retrieves the detailed possible meanings of each mudra, both Samyuktha and Asamyuktha mudras. The sheet will be utilized in this project for obtaining the accurate description.

E. EXPRESSION DETECTION

CNNs have been utilized for facial expression classification of Bharatanatyam performances. CNNs are best known for their capability to learn automatically visual hierarchies of features from visual data, and hence they are found to be extremely efficient for facial expression analysis using subtle differences in visual features. Since Bharatanatyam movements are subtle expressions involving movements of facial elements like eyes, eyebrows, and the mouth, CNNs are ideal to learn such fine patterns. The architecture of CNN is given in Figure 3.

During data-preprocessing, facial expression images were categorized into nine groups and dataset used is in pixel format. Each image was resized to 75×110 pixels with the Python Imaging Library (PIL), converted to black and white to limit input complexity, and converted to NumPy arrays. The unused classes in the dataset were label encoded to create integer classes (0–8) in the rasa. The dataset was divided into two sets, training set and testing set, using stratified sampling to preserve class balance in both.

Three CNN-based models were developed and evaluated: CNN Model 1: Train the unaltered dataset, 39 epochs. The shallow model resulted in overfitting and a testing accuracy of 67% and loss of 1.3. CNN Model 2 (Best Model): Using an augmented dataset, trained for 100 epoch. This model achieved the highest test accuracy of 74%, and lowest test loss of 0.8, it was a model that fit the data well and generalized. Data augmentation included rotating, zooming, horizontal flipping and brightness adjustment for robustness. CNN Model 3: Used transfer learning on the VGG16 architecture with the same augmented dataset. Even with the benefit of using a pre-trained network, it reported slight overfitting with testing accuracy of 58% and a testing loss of 1.3. All models were trained using the Adam optimizer and categorical cross-entropy as the loss function. The input pixel values were normalized before feeding into the models, and the model performance was determined using the test accuracy and test loss metrics. Using comparison analysis, CNN Model 2 was selected for deployment as it produced the best generalization and accuracy from the three models, while balancing depth of model, training time, and dataset augmentation to arrive at an appropriate solution. After the chosen model classifies the dataset, the system generates one of nine possible predicted Navarasa facial expression labels to the input image. To improve explainability, a further mapping function is included in addition, using an externally created Excel spreadsheet as the mapping source with two columns; the Navarasa name, and a semantic description in words. Once a facial expression is classified, the system can then reference the associated meaning from the Excel spreadsheet through a dictionary lookup. Thus, the user has the ability to perform automated recognition of facial expressions and have the contextual meaning associated with each expression, which could be classified as educational or cultural communication. The architecture of the Facial Expression detection is shown in Figure 4. Similar to mudra detection an Excel sheet is

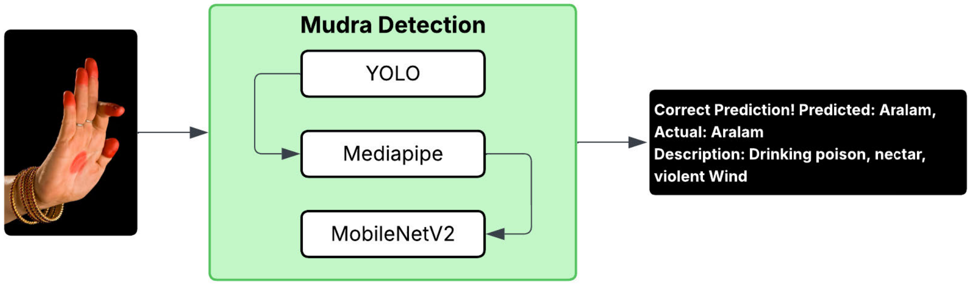

FIGURE 2. Mudra Architecture Diagram (Previous work).

TABLE 2. Descriptions of bharatanatyam mudras.

Mudra	Description
Alapadmam	Viraha (yearning for the beloved), Mukura (mirror), Tataka (pond or lake), Udra-thakopa (great anger), Shakata (cart), fresh ghee, sweets, ball, dancing, palace, cluster of flowers
Aralam	Drinking poison, nectar, Violent wind
Arthachandran	It denotes the eight phase of the waning fortnight of the moon. (Half Moon), A hand seizing the throat, A Spear, Consecrating and bathing an image, A dining plate, source or beginning, Waist, contemplation, Meditation, Mudra for numbers
Arthapataka	Leaves, A board or a slab for writing, Bank of the river, To Indicate "Both", A knife, A banner, A tower, An animal horn
Bramaram	Bee, parrot, wing, cuckoo, heron
Chaduram	Musk, Gold, Copper, Iron, Dampness, to indicate lesser quality, grief, Aesthetic pleasure, eyes, difference in caste, proof, sweetness, slower gait, breaking into pieces, face, oil and ghee
Chandrakala	Moon, face, index of measure, objects of similar shape, The Ganga river, A club (weapon)
Hamsapaksham	Number six, Construction of a bridge, Engraving with nails, To cover
Hamsasyam	An auspicious occasion or festival, Tying thread, Ascertaining the imparted instructions, Horripilation (Romancha), Pearl, Light a lamp, A touchstone (stone meant to test gold), Flowers like Jasmine, To draw picture, Sting, Impress
Kadakhamukham	Plucking or picking flowers, Holding a necklace or a garland, Pulling the bow string, Offer betel leaves, To show preparing a paste of sandal or musk, To mix, To smell, To speak, To see
Kangulam	To represent Lakuca fruit, Bells worn by children, Bell, Chakora bird, Betel-nut tree, The bosoms of young maiden, White Lilly flower, Caataka bird (Skylark), Coconut
Kapitham	Goddess Lakshmi and Saraswati, Milking cows, Holding cymbals, Holding flowers at the time of making love, Grasping the end of the robes, Offering incense or light, Collyrium (applying Kajal)
Kartharimukaham	A Scissors, Separation of a couple, Opposition, Looting, To show two different things, Corner of an eye, Death, Lightning, Sleeping, Falling and Weeping, Creeper
Mayooram	A Creeper, A Bird, Vomiting, Separating the hair locks, Applying Tilak on the forehead, Dispersing water of the river, Something Famous, Discussing the Shastras
Mrigasheersham	A Deer's head, Lord Krishna (when held with both hands), Women's cheek, A Wheel, Fears, Quarrel, Costume or dress, Tripundraka made on the forehead (Tilak on Lord Shiva's forehead), A lute, Massage on the feet, Holding Up
Mukulam	Stepping and calling the beloved
Mushti	Grasping objects, Combative position of the wrestlers, Steady fastness of a person
Padmakosham	Fruit, Woodapple, Bosoms of women, Circular movement, Ball, A vessel, To eat, Bud/ Can be used to show a garland, Mango, Flower strewn around, Cluster of flowers/ bouquet, Offering flowers for worship, A bell, An idol, Ant Hill

created and maintained for Navarasa: The Nine Emotional Expressions in Bharatanatyam as shown in Table 3.

F. LYRICS RECOGNITION & EXPLANATION

1) AUDIO-TO-TEXT CONVERSION USING WHISPER

To accurately extract the song lyrics present in the Bharatanatyam performance videos, explorations were conducted on a variety of active speech recognition (ASR) models, including Google Speech-to-Text API and Whisper (small, medium, and large). The project evaluated a number of Whisper models to assess their success in segmenting Bharatanatyam performance song lyrics. A comparative analysis of these model is summarized in Table 4, highlighting difference in word error rate (WER), processing time, language support, and overall accuracy. There were challenges in transcription in the base language, particularly

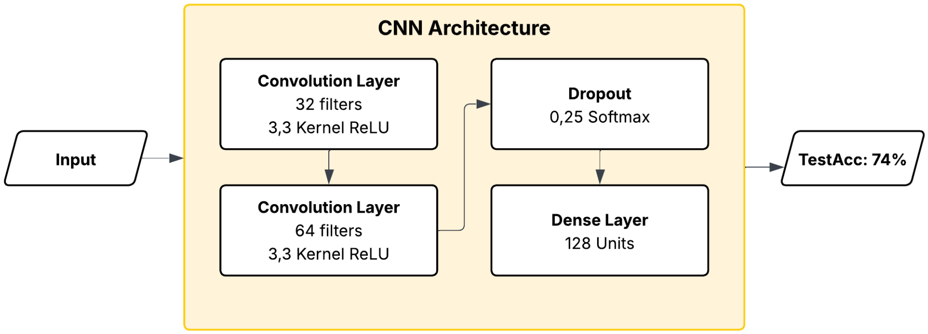

FIGURE 3. Convolutional Neural Networks (CNNs) Architecture.

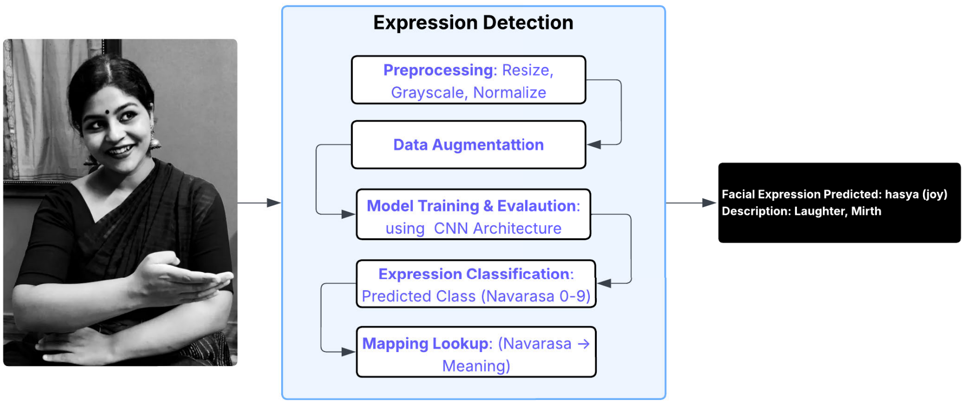

FIGURE 4. Expression detection model architecture.

TABLE 3. Navarasa: The nine emotional expressions in bharatanatyam.

Name	Meaning
Srngara (Love)	Love, Beauty, Romance
Hasya (Joy)	Laughter, Mirth
Karuna (Sorrow)	Sorrow, Compassion
Raudra (Anger)	Anger, Fury
Vira (Confidence)	Courage, Heroism
Bhayanaka (Fear)	Fear, Terror
Bibhatsa (Disgust)	Disgust
Adbhuta (Wonder)	Astonishment, Surprise, Wonder
Santa (Peace)	Peace, Tranquility

for Tamil songs due to phonetic variation and language complexity as a whole. Of the models evaluated, the Whisper Large v2 model was conclusively the most accurate identifier and transcriptor of lyrics. The Whisper Small model was quite computationally efficient, but it struggled with phonetic variations in Tamil songs and had frequent misidentifications. Whisper Medium was an intermediary between efficiency and accuracy, but did not adequately capture the poetic nature of classic Tamil lyrics when set to background music. Whisper Large model performed better in accuracy and recognition of different languages; however, part of the issue remained in the transcription of Tamil language words that sounded similar and the context was not generally accurate. After significant testing of the various models, the final model selected for the project was decided to be the whiper large v2 model based on accuracy, phonetic detail, and fewest transcriptions in cases of throttling. Once it was established that whiper large v2 offered the least amount of errors in capturing original lyrics, it was determined to be the chosen ASR model for extracting lyrics. This model was the most effective for extracting lyrics from videos of Bharatanatyam performances with longer next-lined lyrical content in the video while keeping the original phonetic details. Like other ASR models, among other disadvantages,

TABLE 4. Comparison of ASR models for bharatanatyam lyrics transliteration.

Model	Version	WER (%) ↓	Time (s) ↓	Language Support	Accuracy (%)
Whisper Base	v1	28.5	2.1	Multilingual	70.2
Whisper Medium	v1	18.9	3.7	Multilingual	80.4
Whisper Large	v1	12.3	5.5	Multilingual	89.1
Whisper Large	v2	10.1	5.4	Improved low-resource	**91.6**
Whisper Large	v3	**8.7**	6.2	Enhanced Indian support	65.0
Google ASR (Cloud)	—	14.2	2.9	Multilingual (Cloud)	50.7

the challenge remained within using whiper large v2 and processing the pronunciation since it was highly dependent on the modulation of the classical music within the presentation. However, it provided the most accurate and structured transcripts for ASR from the models compared.

2) SEMANTIC ANALYSIS AND TRANSLITERATION USING GEMINI

To analyze the lyrical content contained within Bharatanatyam performances, we used Gemini, which is a leading LLM, to accomplish two key tasks: transliteration and then semantic analysis. The original song lyrics, typically in Indian classical or regional languages (for example, Tamil, Sanskrit), were first entered into Gemini for script-level transliteration into Roman script (with the intent of having phonetic consistency in the pipeline and honouring some of the language ties). Then, we used Gemini to conduct zero-shot semantic extraction, where Gemini was used to generate natural language bounds on the lyrical content in English. This significant aspect is that these bounds attempted to encapsulate the contextual meaning, metaphors, and cultural symbolism taken from the original lyrics. After we created the interpreted output, we were able to integrate this with the multimodal module of the system, including mudra classification and facial expression classification, to enable some cross-modal alignment and in the end enable semi-automated narrative reconstruction of the dance-piece.

G. MULTIMODAL INTEGRATION AND NARRATIVE GENERATION USING GEMINI

This comparative evaluation of leading prompting platforms ChatGPT, Gemini, and Groq API is deemed useful in determining the best prompting platform for our system. The comparison presented in Table 5 considers various factors, including provider, costs, accuracy of outputs, handling capabilities of prompts, multimodal, context length, speed, and token numbers. Of the platforms considered, we selected Gemini for our implementation, due to its accuracy, multimodal capabilities, a context window of up to 1 million tokens, and, cost, providing good value for use case.

Gemini was chosen for multimodal semantic fusion owing to its rich functionality, large-context reasoning capacity, and cost-effective implementation. Its multimodal features enable the integration of gestures, emotions, and lyrics into cohesive narrative interpretations. Future extensions will include a Retrieval-Augmented Generation (RAG) mechanism using traditional sources such as the Natya Shastra [36] and Abhinaya Darpanam [37] to enhance cultural alignment and authenticity.

To generate a coherent narrative that represented the underlying story of a Bharatanatyam performance, we interfaced with Gemini, a multimodal semantic integration system that is a LLM system. The overall system is comprised of three separate descriptive inputs: (i) the song lyrics transliterated and interpreted, (ii) the mudra classification output along with semantic descriptions of the meaning of the mudra, and (iii) the emotional analysis of facial expressions and associated emotional descriptors. The outputs of the three descriptive layers were pre-processed and saved into a single description.txt file so that the text could be accessed programmatically during runtime for reading/writing.

The interaction with Gemini was based on prompt designing. The description files were concatenated with a structured template and read as a single prompt. Gemini was instructed to create a high level textual storyline that represented the narrative that was intended to be inferred through the combined visual (mudra and expression) and auditory (lyrics) modalities. The system aligns context and time across all interpretative layers and allows for the automatic generation of human-readable dance interpretations that preserved both the emotion and semantic content of the dance performance.

H. OUTPUT GENERATION: TEXT AND AUDIO STORYLINES

After generating the final narrative storyline using Gemini based on inputs from lyrics, mudras, and facial expressions, the output is exported in two formats:

1) PDF EXPORT OF NARRATIVE STORY

The storyline developed as text by Gemini is programmatically saved as a PDF via Python libraries such as FPDF or ReportLab without the participation of Gemini, involved through customized scripts advancing documentation, storage, and sharing of the interpreted dance story.

2) AUDIO GENERATION USING TTS

To develop an audio version of the storyline, Gemini was used to create voice-friendly narrative prompts that were fed to internal text-to-speech (TTS) processing to create a natural audible version of the storyline (e.g..mp3 or.wav). The audio format improves accessibility and creates an immersive storytelling experience for users.

I. SYSTEM OVERVIEW AND SUMMARY

The proposed system utilizes AI and multimodal integration to obtain interpretations of Bharatanatyam performances by analyzing mudras, facial expressions, and lyrics in spoken and sung languages. The proposed model used YOLO and MediaPipe to detect and classify mudras in video, and since the data is not sequence-dependent, a MobileNetV2transfer learning model was employed. Similarly, facial expressions were analysed with a set of trained CNN-based models based on Navarasa's nine natural feelings, or expressions. Lyrics

TABLE 5. Comparison of prompting platforms: ChatGPT vs Gemini vs Groq API.

Feature / Criteria	ChatGPT (GPT-4)	Gemini (1.5 Pro)	Groq API
Provider	OpenAI	Google AI	Groq
Cost	High (token-based billing)	Free or affordable tiers	Free
Output Accuracy	Excellent	Excellent	Good (model-dependent)
Prompt Support	Advanced chat and function calling	Advanced prompting with context tools	Raw prompt only
Multimodal Support	Yes (text, image, audio)	Yes (text, image, audio, video)	Text-only
Context Size	128k tokens (GPT-4o)	Up to 1M tokens	∼65k tokens (model-dependent)
Speed	Fast	Moderate	Ultra-fast inference
Token Limit Range	128k	Up to 1M	∼65k
Reason for Not Choosing	Expensive at scale	—	Limited context structure
Why We Used This	—	Correct output, low cost, rich features	—

were transcribed using the Whisper Large v2 Model, which accurately recognizes phonetics, including Tamil.

Gemini was employed to transliterate the lyrics into Roman concepts and for generating semantically rich representations. The final elements of mudra and facial expression classification and the contours from semantics were collated into a single source description.txt file. A structured prompt file was constructed using the XML structure from facial expression recognition to merge the mudra, facial expressions, and descriptions and passed to the Gemini model to generate the final narrative storyline. A very unique aspect of the narrative was that it was exported into two formats, a programmatic PDF file which focused on documentation, and a audio file output, generated through text-to-speech synthesis, to enhance accessibility and engagement, and tell an engaging story. A brief practitioner check was carried out with trained Bharatanatyam performers, who reviewed the mudra and rasa outputs to confirm their cultural appropriateness.

IV. EVALUATION

This section presents the experimental setup, evaluation methods, and results obtained from implementing the proposed Bharatanatyam interpretation system. The goal of the experiments was to assess the effectiveness of individual modules—namely mudra recognition, facial expression classification, lyric transcription and interpretation—as well as the overall performance of the multimodal fusion approach for storyline generation. Each module was tested independently using curated datasets and then integrated into the complete pipeline for end-to-end evaluation. The mudra recognition model was trained using hand landmark keypoints extracted via MediaPipe and classified using a MobileNetV2 architecture. Facial expression analysis was carried out using CNNs trained on a dataset labeled according to the nine Navarasa categories. Whisper Large-v2 was used to transcribe audio lyrics from Bharatanatyam videos, and Gemini was employed for both transliteration and semantic interpretation of the lyrics. The final output a narrative storyline was generated through prompt-based interaction with Gemini using a structured input file containing mudra, expression, and lyric interpretations. This output was then exported in both textual (PDF) and audio formats. The effectiveness of each module and the quality of the final narrative were evaluated qualitatively and quantitatively, and the results demonstrate the system's capability to produce coherent, culturally relevant interpretations of Bharatanatyam performances.

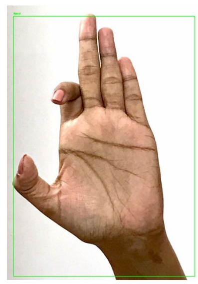

FIGURE 5. Object detection using YOLO.

A. MUDRA IDENTIFICATION

The dataset has a rich set of Bharatanatyam mudra images, all preprocessed with care to maximize the visibility of the hands. To start the pipeline, the YOLO object detection model was used to localize the areas of the hand, even against visually cluttered or complex backgrounds. As shown in Figure 5, YOLO was able to detect hand regions that served as the foundation for the mudras with high accuracy irrespective of varying image conditions.

After detecting the hand areas, the regions were processed with MediaPipe, a landmark detection system with hand keypoint detection optimized for this purpose. MediaPipe output 21 individual keypoints per hand that recorded the anatomical structure consisting of the wrist, finger joints, and tips, as illustrated in Figure 6. The spatial features were critical in discriminating visually ambiguous mudras, like Aralam, from other mudras.

Next, the hand keypoints that were extracted were fed into a MobileNetV2-based classifier. During this phase of

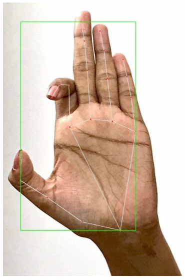

FIGURE 6. Key points detection by using mediapipe.

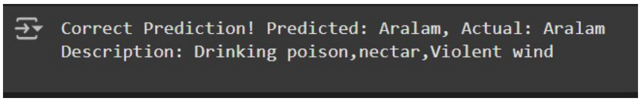

FIGURE 7. Output of mudra detection.

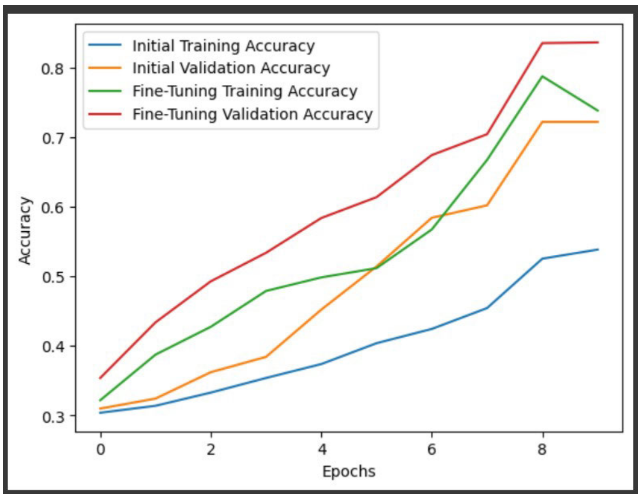

FIGURE 8. Accuracy vs epochs graph.

the pipeline, the respective mudra label was predicted with strong confidence—correctly classifying the Aralam mudra in this case. For additional interpretability, a hand-collected database maintained in a Microsoft Excel document was queried to fetch descriptive data about the found mudra's symbolic and cultural context. The ultimate output displayed on-screen was the name of the mudra and its original meaning, as in Figure 7.

The fine-tuning of the classification model improved its performance significantly compared to the baseline. Of interest is how the fine-tuned model achieved over 83% validation accuracy at the eighth epoch, outperforming the base model and with superb generalization with zero overfitting. Figure 8 is a plot of how the accuracy of the model evolved during training.

B. FACIAL EXPRESSION

This section presents the results of facial expression classification based on the Navarasa theory in Bharatanatyam. CNNs were employed to analyze facial cues and predict the corresponding emotional category from video data.

1) DATASET PREPROCESSING

Facial images were preprocessed by resizing them to 75 × 110 pixels, converting them to grayscale using the Python Imaging Library (PIL), and storing them as NumPy arrays. The dataset was divided into nine classes corresponding to the Navarasa expressions, with unused classes removed and label-encoded to integer values ranging from 0 to 8. Stratified sampling ensured class balance in both training and testing sets.

2) MODEL TRAINING AND EVALUATION

Three CNN models were developed and trained using the Adam optimizer and categorical cross-entropy loss. Input pixel values were normalized for all models.

Model 1 (Baseline CNN): Trained on the original dataset for 39 epochs. It exhibited overfitting, resulting in a testing accuracy of 67% and a test loss of 1.3.

Model 2 (Best Performing CNN): Trained on an augmented dataset for 100 epochs. Data augmentation techniques included image rotation, zooming, brightness adjustment, and horizontal flipping. This model achieved the highest test accuracy of 74% with the lowest test loss of 0.8, demonstrating strong generalization capability. The training and testing accuracy across epochs is shown in Figure 11, and the corresponding loss trends are also visualized in the Figure 11.

Model 3 (Transfer Learning with VGG16): Utilized the pre-trained VGG16 architecture on the same augmented dataset. Despite transfer learning, the model showed slight overfitting, with a testing accuracy of 58% and test loss of 1.3.

A sample output of facial expression prediction is illustrated in Figure 9, and the confusion matrix for Model 2, highlighting class-wise performance, is shown in Figure 10.

3) EXPRESSION INTERPRETATION AND MAPPING

The best model (Model 2) is chosen for deployment. The model predicts the one of the nine Navarasa categories for each prediction. So, with this use case in mind, an Excel file which contains two columns, Navarasa name, and semantic description, to increase interpretability was developed, and was used for mapping. Once a facial expression is classified, a dictionary lookup provides the associated cultural and emotional meaning. This final output not only allows for automated inference of facial expressions, but also provides meaning and contextualizes the emotional experience of the facial expression that increases the educational and cultural relevance of this system.

C. LYRICS EXTRACTION

To reliably identify languages, and transcribe lyrics to Bharatanatyam songs, we opted to use the whisper large v2 model developed by OpenAI. This state-of-the-art

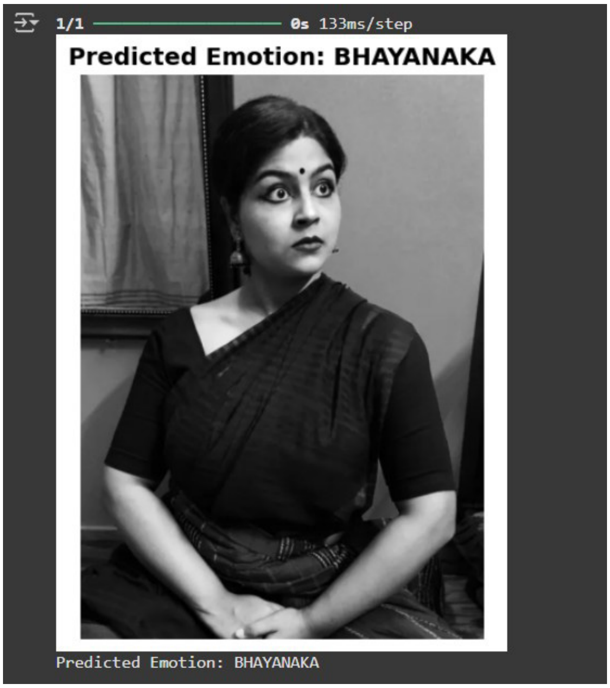

FIGURE 9. Facial expression output.

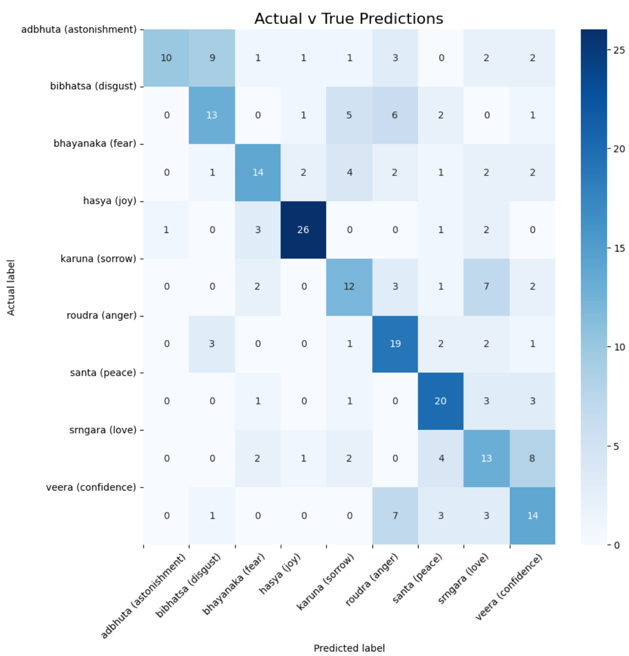

FIGURE 10. Confusion matrix highlighting class-wise performance of Facial Expression Prediction.

multilingual speech recognition system can recognize over 90 different languages and was tested using classical Bharatanatyam works along with media. The model produced recognized Tamil, Telugu, and Sanskrit languages and transcribed lyrics with proper words in context and accurately. It showed resilience to variations in pronunciation, recording environments, or background noise. These transcriptions, as illustrated in Figure 12, were essential for linking the lyrical content to the related mudras and facial expressions, enabling the extraction of richer semantic meaning from the dance. These results validate Whisper's potential for further development in the linguistic analysis of classical performance contexts.

D. SEMANTIC INTERPRETATION OF LYRICS USING GEMINI

Gemini, a LLM, was utilized to transliterate and interpret Bharatanatyam song lyrics. The process involved:

1) TRANSLITERATION

Lyrics in Tamil/Sanskrit were accurately converted into Roman script, maintaining phonetic integrity.

Prompt Used for Transliteration:

You are an expert in Tamil-to-English transliteration.

The following text is a line from a Tamil devotional song:

`{lyrics}`

Only the transliterated part is needed — do not give translation or explanation. Output should be in simple Romanized Tamil.

Output:

Alaipayuthe Kanna en manam alaiapayuthe Ananda Mohana venuganamathil alaiapayuthe Kanna en manam alaiapayuthe unnAnanda Mohana venuganamathil alaiapayuthe Kanna.

2) SEMANTIC EXTRACTION

Gemini was then used to extract the contextual and cultural meaning from the transliterated lyrics using zero-shot prompting.

Prompt Used for Gemini:

You are an expert in song analysis. Please provide a brief, single-paragraph explanation of the meaning and themes of the following Tamil devotional song in the easiest and simplest way for common people to understand.

`{transliterated_lyrics}`

Describe the main emotional and spiritual themes, including any religious or philosophical context, in a concise and informative manner. Focus on the core message and the significance of the song.

Output:

While this devotional song in Tamil is presented as a romantic appeal to "Kannā" (a term frequently associated with Krishna, the Hindu deity), it is meant to be more than simple romantic love. It gives an allegorical view of the soul yearning for union with the divine. The "lovesick" identity characterizes a person who is seeking his or her spiritual liberation very badly ("Ananda Mohana" are blissful liberation) after he or she feels like he or she has lost the divine identity. The urgency and intensity of the love expressed indicates the level of desperation desired to be in union with ultimate reality. The reconciliation is more than a classic romantic relationship. It is a spiritual awakening that illustrates that the Bhakti (devotional) tradition is idealizing

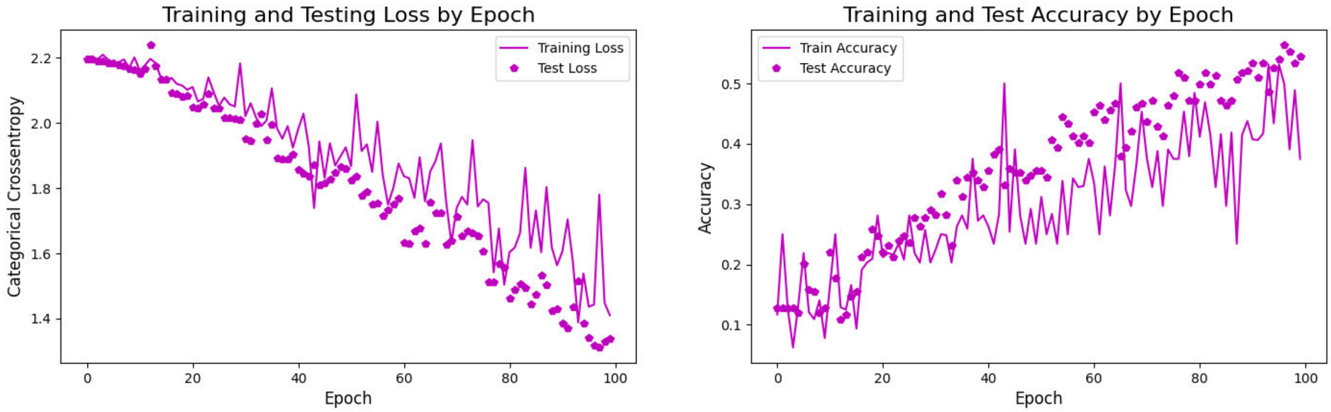

FIGURE 11. Training and testing accuracy by epoch & Training and testing loss by epoch.

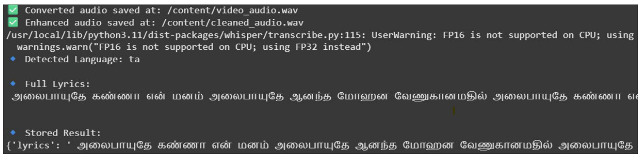

FIGURE 12. Whisper output for lyrics transcription.

human love in terms of the soul's devotion to God. The essence of this song demonstrates the transformative quality divine love has along the process of the deeply emotional journey of attaining spiritual fulfillment.

E. NARRATIVE GENERATION USING LLM (GEMINI)

The Gemini LLM was used for multimodal semantic integration to yield a coherent and meaningful narrative that embodied the underlying story of a Bharatanatyam performance. The system integrates three separate inputs: Interpreted song lyrics Mudra classification predictions and their semantic meanings Facial expression recognition predictions and the associated emotion descriptors Each input was preprocessed and placed in a single massively unified description.txt file. During runtime, this file was accessed programmatically that allowed the LLM to read and write the needed content - thus allowing us to form the necessary prompt. Through prompt-based interaction, all three inputs were joined together in a templated manner, and entered into Gemini as a single prompt. The LLM was given an instruction to create a high-level narrative that captured the intended meaning and emotional spirit conveyed by the unified audio-visual input. The output was a well-formed textual narrative that showed contextual cohesion and alignment between the lyrics, gestures (mudras), and expressions of the facial expressions. This narrative also adequately reflected the semantic significance and emotional spirit of the performance - showing that LLMs like Gemini are capable of automation, classifier interpretation of classical dance forms.

F. INTERPRETATION DATA

1) SONG DESCRIPTION OBSERVED

This Tamil devotional composition is framed as a romantic plea to "Kannā" a name traditionally associated with the Hindu deity Krishna. However, the song's core message transcends romantic sentiment. The central figure in the song—a lovesick devotee—symbolizes the individual soul yearning to reunite with the divine. The phrase "Ananda Mohana" refers to the blissful liberation that one experiences upon spiritual awakening. The anguish and longing embedded in the lyrics reflect the emotional turmoil of separation from divine consciousness. This intense yearning ultimately becomes a metaphor for the Bhakti (devotional) tradition's core philosophy: the use of human love to depict the soul's quest for union with God. The emotional progression in the song captures the transformative nature of divine love, illustrating how longing can evolve into spiritual realization.

2) FACIAL EXPRESSIONS OBSERVED

- **Facial Expression:** `santa (peace)`
  Description: Peace, Tranquility
- **Facial Expression:** `srngara (love)`
  Description: Love, Beauty, Romance
- **Facial Expression:** `hasya (joy)`
  Description: Laughter, Mirth
- **Facial Expression:** `adbhuta (astonishment)`
  Description: Surprise, Wonder

3) MUDRAS OBSERVED

- **Mudra:** `kadakhamukham`
  Description: Depicts actions such as offering a garland, plucking flowers, or preparing sandal paste.
- **Mudra:** `mayooram`
  Description: Symbolizes a creeper, a peacock, applying tilak, or referencing something graceful or scriptural.
- **Mudra:** `hamsasyam`

Description: Associated with auspicious occasions, drawing, string-tying, or signifying delicate objects like jasmine flowers or pearls.

- **Mudra:** `alapadmam`
  Description: Indicates yearning (viraha), ponds, flowers, and emotional states such as anger or longing.

4) NARRATIVE GENERATION

Prompt Used for Generate Storyline:

You are a Bharatanatyam expert AI. The following text contains: - Bharatanatyam mudras and their meanings - Facial expressions and their meanings - The meaning of a song

Your task is to read this data and write a single, continuous **storyline** that connects the mudras, facial expressions, and song meaning together. The story should reflect what the dancer is expressing and how the movements and emotions tell the story. Write it in elegant, easy-to-understand English.

Here is the content:

`{input data}`

Now generate the storyline.

5) GENERATED STORYLINE OUTPUT

Generated Narrative Interpretation:

The dancer, bathed in the soft light of the stage, begins with a serene expression of santa (peace). Her hands form the alapadmam mudra, initially suggesting the deep viraha (yearning) of the soul separated from the divine. The alapadmam also hints at a still lake (tataka), reflecting the vastness of the spiritual journey ahead. Her face, though peaceful, carries a subtle undercurrent of longing. The song begins, its melody echoing the soul's quiet desperation for union with Kannā. As the song builds, so does the dancer's emotion. The srngara (love) blossoms across her face, a passionate yearning for the divine reflected in the delicate kadakhamukham mudra, depicting the offering of a garland – a symbolic gesture of devotion. She seems to pluck unseen flowers, each movement a prayer. A flicker of hasya (joy) crosses her lips as she imagines the eventual reunion, a fleeting glimpse of the blissful ananda mohana promised in the song. Suddenly, her expression shifts to adbhuta (astonishment). The mayooram mudra unfolds, suggesting the spreading of a creeper – a metaphor for the divine grace that slowly envelops the devotee. The unexpected wonder reflects the profound transformation hinted at in the song's message. The dancer's body language expresses the overwhelming experience of divine love, a grace that disperses doubts and cleanses the soul. The hamsasyam mudra, with its multiple interpretations, comes into play, signifying both the auspiciousness of this spiritual awakening and the difficulty of the journey - the drawing of a picture of enlightenment alongside the piercing sting of separation. The dance culminates in a powerful expression of devotional love. The initial viraha is replaced by a profound sense of gratitude and awe, her face radiating the joy of union. Though the specific mudras might shift, each movement and expression remains consistent with the song's powerful message of the soul's transformative journey towards spiritual liberation, culminating in the ultimate reunion, not as romantic love but as the blissful union with the divine. The dancer embodies the soul's fervent plea, its struggles, and its ultimate triumph, making the allegorical narrative of the song vibrantly real.

V. ERROR ANALYSIS

This section presents the major errors and issues experienced while developing and testing the NATYA-AI system. Despite the system generating encouraging results, some major limitations were found that affected its accuracy and reliability.

A. MUDRA RECOGNITION ERRORS

The mudra recognition component utilized YOLO for hand detection, MediaPipe for keypoint extraction, and MobileNetV2 for classification. Although the system had a near 83% accuracy, misclassifications were observed—especially among visually comparable mudras such as Aralam and Tripataka. These were caused by scant training data, inadequate lighting conditions in input videos, or hand position overlaps. As Bharatanatyam hand gestures are subtle and intricate, lack of explicit temporal modeling also reduced accuracy. Improved development can be achieved with LSTM or 3D CNNs to capture gesture sequences more effectively.

B. DATASET LIMITATIONS

One of the primary challenges that were encountered in this research was that there were no publicly available mudra datasets for Bharatanatyam. This greatly hindered learning and generalization capabilities of the model. In order to address this issue, we generated a bespoke mudra dataset consisting of images from both single-hand and double-hand gestures. The dataset was manually labeled and structured to facilitate proper training and classification. Although this improved the performance of the model, a bigger and more extensive dataset will still be required for wider usage.

C. FACIAL EXPRESSION DETECTION ERRORS

The facial expression classification model, learned on the Navarasa framework with CNNs, achieved a test accuracy of 74%. But there were misclassifications between expressions such as Hasya (joy) and Adbhuta (wonder) when the expression of the dancer was subtle or half-covered. Accuracy was influenced by aspects such as lighting, camera view, and ornamentation. Attention-based models or keypoint-based expression analysis can be used to enhance this in future research.

D. LYRICS TRANSCRIPTION ISSUES

The Whisper Large v2 model was able to transcribe Tamil lyrics even with background music. Misinterpretations still existed due to poetic pronunciation, musical modulation, and vocally overlapping utterances. These misinterpretations had minimal effects on the semantic meaning of the lyrics. Adjusting the model for classical music or using sophisticated audio preprocessing techniques would enhance transcription accuracy.

E. STORYLINE GENERATION CHALLENGES

Gemini, the big language model, was used to generate interpretive narratives from mudras, facial expressions, and lyrics. The majority of the narratives were well-coherent, with some outputs lacking adequate alignment between gesture and lyrical/emotional content. This was mainly due to input data gaps or lack of consistency. Better prompt design and structured input forms can guarantee fewer alignment issues in the future.

F. OVERALL SYSTEM LIMITATIONS

The system performance overall is now constrained by the size and variety of the data. The system does not carry out real-time processing yet, which restricts its use in live dance performances or interactive learning. To better capture micro-expressions and transitional movements, we plan to integrate temporal encoders (e.g., LSTM/Transformer-based video modules). A larger dataset with more dancers, age groups, styles, and performance contexts will be collected to improve robustness and reduce bias.

VI. DISCUSSION

This study introduces NATYA-AI, a multimodal system developed for the semantic interpretation of Bharatanatyam performances through the integration of computer vision techniques, speech recognition, and LLMs. The experimental results highlight the potential of such an approach while also revealing challenges associated with the application of AI to traditional, symbolically rich cultural practices.

Unlike prior multimodal frameworks that primarily combine sensory inputs for generic perception tasks, *NATYA-AI* achieves knowledge-aligned fusion by embedding cultural semantics of *mudras*, *Navarasa*, and devotional lyrics, thereby transforming off-the-shelf AI components into an interpretable narrative system for traditional performing arts.

The mudra recognition module, which employs YOLO and MediaPipe for hand region detection and MobileNetV2 for classification, achieved an accuracy of 83%. This performance demonstrates the system's ability to identify intricate hand gestures under diverse background and lighting conditions. However, the model currently relies on frame-wise static classification, which limits its ability to interpret sequential transitions between gestures. Incorporating temporal models such as Long Short-Term Memory (LSTM) networks or transformer-based video encoders could enhance the system's capability to model gesture sequences, improving narrative continuity and temporal context alignment.

The facial expression classification component, based on CNNs trained on augmented Navarasa datasets, yielded a maximum test accuracy of 74%. Although this indicates reasonable generalization performance, the system exhibits reduced sensitivity in differentiating closely related expressions, particularly those involving fine-grained movements of the eyebrows and eyes. Hybrid approaches that combine CNNs with recurrent or attention-based layers may improve detection of these subtleties by capturing both spatial and temporal dependencies in expression dynamics.

Whisper Large-v2 was employed as the backbone for the audio transcription module and performed well under varied acoustic conditions, including classical Tamil audio with musical accompaniment. Nevertheless, transcription quality was occasionally impacted by background instrumentation and the complexity of poetic constructs. Future iterations of NATYA-AI may incorporate fine-tuned automatic speech recognition (ASR) models specifically adapted to Indic languages to improve phonetic representation and semantic fidelity in lyrical contexts.

The semantic interpretation component, built using the Gemini LLM, effectively mapped transcribed lyrics to culturally contextualized meanings. When integrated with mudra and expression recognition outputs, this module was capable of generating coherent interpretive narratives aligned with the thematic intent of the performance. However, to improve alignment consistency in longer or more complex sequences, future designs may benefit from the integration of multimodal embedding frameworks or transformer-based fusion architectures, enabling more structured and context-aware narrative synthesis.

One practical feature of the system is its ability to generate dual-format outputs—PDF for text-based narratives and audio for auditory accessibility. These outputs are designed to address diverse user needs, including those of visually impaired individuals or learners unfamiliar with classical Indian languages. This inclusive design aligns with Sustainable Development Goals (SDGs) concerning equitable education, cultural inclusion, and digital preservation of intangible heritage.

Despite the encouraging results, the system currently operates with limitations stemming from dataset constraints. The training data includes a limited range of performers, lighting conditions, and regional variations in style. To improve generalization and robustness, it is essential to expand the dataset with broader demographic and contextual diversity. Furthermore, enabling real-time inference would extend NATYA-AI's applicability to live performance analysis, interactive learning environments, and classroom demonstrations.

Although the current implementation focuses on the recognition and fusion of gestures, facial expressions, and lyrical meaning, the design of *NATYA-AI* conceptually aligns with the performative grammar described in the *Natya Shastra*. The visual (*mudra*) module corresponds to the Āṅgika *abhinaya* (expression through the body), the audio–lyric component to Vācika *abhinaya* (expression through speech), and the facial emotion analysis to Sāttvika *abhinaya* (expression of inner emotion). The fusion of these modalities aims to approximate the holistic process through which *rasa* (aesthetic essence) is evoked in Bharatanatyam. While the present work does not yet formalize the complete symbolic rules of the *Natya Shastra*, its modular architecture provides a foundation for incorporating *rasa*-level mapping and symbolic reasoning in future extensions, advancing computational models that remain faithful to classical performance theory.

Novelty and Cultural Grounding: The novelty of *NATYA-AI* lies in its integration of multimodal AI with cultural semantics, uniting computer vision, speech recognition, and large language modeling within the framework of classical Indian aesthetics. The system achieves semantic coherence by mapping gestures, facial expressions, and lyrical meaning to culturally defined constructs of *abhinaya* and *rasa*, guided by canonical texts such as the *Natya Shastra* [36] and *Abhinaya Darpana* [37]. The dataset and evaluation components have been expanded to report performance metrics for *mudra* and expression recognition, as well as qualitative coherence in narrative synthesis. This alignment of cultural theory with multimodal fusion and quantitative evaluation establishes *NATYA-AI* as both a technically rigorous and culturally faithful framework in the domain of digital heritage.

One of the limitations of the present study is in the limited dataset size and diversity. The future work would involve systematic dataset growth with the inclusion of recordings by various performers, local dance styles, and multilingual songs. We plan to build a publicly available benchmark dataset, which will allow for strong training and unbiased comparative evaluation of Bharatanatyam interpretation models.

In conclusion, NATYA-AI provides a proof of concept for the automated semantic interpretation of classical Indian dance using multimodal AI. The architecture demonstrates technical feasibility while also highlighting areas for refinement, including model robustness, cross-modal integration, and real-time deployment. The combined analysis highlights how individual modules complement each other, especially when gesture or expression predictions are uncertain. These findings underscore the utility of AI in enhancing interpretive access to traditional art forms and contribute to ongoing efforts in Cultural AI for heritage documentation, education, and outreach.

A. CULTURAL VALIDATION AND PRACTITIONER FEEDBACK

A key part of assessing NATYA-AI is to ensure its interpretations are culturally authentic and pedagogically aligned. One of the co-authors of this paper is a trained Bharatanatyam (BN) dancer, whose skills helped determine the mapping of mudras to emotions and the semantic construal of the meaning of the lyrics. The practitioner perspective has provided a first stage of validation for whether the textual outputs of the system reflect the symbolisms and intention of expressiveness of the tradition.

In future work we will continue this validation with structured feedback sessions consisting of Bharatanatyam teachers, scholars and advanced students who will be asked to evaluate cultural fidelity, pedagogical worth, and usefulness in the context of practice and performance situations. So far, we have focused on practitioner feedback to inform modification of the system, resist cultural emptiness simplifications and to aligning NATYA-AI with the practitioner dance community and the audience at large.

B. ETHICAL, CULTURAL, AND ACCESSIBILITY CONSIDERATIONS

Using AI on a traditional and sacred form of art like Bharatanatyam requires careful attention to moral, cultural, and social sensitivities. NATYA-AI is framed as an assistive framework that serves to support human artistry and teaching, not to take its place. The outputs of NATYA-AI are designed to assist learners, researchers, and all audiences without compromising the spiritual, symbolic, and performative powers of the tradition. By performing the translations of mudras, expressions, and lyrics into narrative words, cultural integrity is retained because they are understood in their proper narrative contexts. NATYA-AI is intended to work collaboratively with practitioners and cultural scholars to inform interpretation, validate mappings, and to represent intentional meanings as anything less runs the risk of falling into shallow or simplistic understandings of symbolically laden meanings. It is hoped that through NATYA-AI's various outputs, which visually narrate each sequence in text and audio formats, accessibility is broadened for persons who are blind or low vision, audiences who may be non-native, and audiences who may not be fluent in Indic languages. Further developments will include interactive learning modules designed especially for students, educators, and international audiences. Collectively, these responses model ethical approaches of fairness, transparency, and accessibility in developing Cultural AI.

VII. CONCLUSION

This paper presented **NATYA-AI**, a multimodal architecture designed for the semantic interpretation of Bharatanatyam performances by integrating gesture recognition, facial emotion classification, and lyrical transcription with LLM-based reasoning. The system is specifically tailored to address the interpretive challenges posed by classical Indian dance forms, which rely heavily on symbolic, non-verbal communication.

NATYA-AI differentiates itself from conventional video summarization or speech transcription tools by focusing on cultural semantics and narrative cohesion. It successfully identifies and classifies mudras and facial expressions aligned with the Navarasa framework, transcribes multilingual devotional lyrics using Whisper ASR, and synthesizes these inputs using large LLMs to generate structured textual and audio storylines. These outputs contribute to improved accessibility, particularly for learners, researchers, and non-native audiences, and serve as supplementary material for educational and archival purposes. Future expansions of NATYA-AI will emphasize dataset diversity, such as increased participation of performers, diverse performance settings, and multilingual repertoire.

The modular architecture of NATYA-AI supports future scalability and integration. Proposed extensions include real-time feedback mechanisms for performance evaluation, the incorporation of other Indian classical dance traditions to increase generalizability, and the development of interactive user interfaces to enhance engagement and usability. The system design also allows for incremental improvements, such as fine-tuning the models on diverse datasets and integrating multimodal transformer architectures for enhanced context alignment. This work thus establishes NATYA-AI as a novel, knowledge-guided multimodal framework that unifies vision, speech, and language models to semantically interpret and preserve the narrative essence of Bharatanatyam.

In conclusion, NATYA-AI demonstrates the feasibility of applying deep learning and natural language processing techniques to culturally embedded performance arts. Its contribution lies not only in technical implementation but also in supporting broader goals of digital heritage preservation and inclusive cultural education.

In future work, we plan to extend NATYA-AI with a domain ontology and Retrieval-Augmented Generation (RAG) layer to encode Bharatanatyam's formal grammar—linking gestures, emotions, and narrative rules for deeper, knowledge-guided interpretation. Thus we will expand the semantic layer by incorporating structured rules from the Natya Shastra and Abhinaya Darpanam for deeper symbolic grounding. The framework aligns with ongoing research in Cultural AI and presents a practical foundation for further interdisciplinary advancements in computational ethnography, performing arts analytics, and accessible learning systems. Future work will focus on optimizing the pipeline for low-latency processing so that NATYA-AI can support real-time classroom or stage demonstrations.

VIII. CODE AND RESOURCE AVAILABILITY

The source code, model weights, and example data for the NATYA-AI system are available at:

`https://github.com/GeethikakpkNambiar/NATYA-AI`

The repository includes scripts for mudra and expression classification, along with sample input videos (short clips), transcription results, and narrative generation outputs. While representative videos are provided, the full set of extracted unique frames is not included due to storage constraints. Nevertheless, the included samples are sufficient to reproduce and understand the core functionality of the NATYA-AI system.

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